# CSE 219 — Signals and Linear Systems

## Lec 7 · Fourier Transform: Pairs, Properties, Filters

*Lesson notes · Exam approach · TF practice with solutions · Quick notes (Anik Sir's portion)*

## How to Read These Notes

| | |
|---|---|
| **EXAM FOCUS** | Exam importance, where it appears in the TF collection |
| **MEMORIZE** | Rules, tables and formulas to memorize |
| **TRAP** | Common mistakes and exam traps |
| **EXAM-READY ANSWER** | Model wording you can write in the exam |
| **NOTE** | Extra explanation / clarification |
| **Grey box, monospace** | Code, worked tables, derivations and hand-drawn sketches |
| **Centered grey box** | Formula to remember / final numerical result |
| **Purple heading / ✓ green** | Practice questions / their solutions |

## Contents

**Exam Map — What the TF Papers Ask** — Fourier-transform questions and their weight

**Lesson 16 — From Fourier Series to Fourier Transform** — T → ∞, the transform pair, existence, FT of periodic signals

**Lesson 17 — Rectangles, Shifts and Modulation** — rect ↔ sinc, time shift, frequency shift, cos-modulation, δ

**Lesson 18 — Transform Pairs and Properties Toolbox** — $e^{-at}u$, sign, u, scaling, duality, d/dt, ∫, convolution, Parseval

**Lesson 19 — LTI Systems in Frequency: Filters, Distortionless, Bandwidth** — H(ω), ideal filters, echo filter, PSD, 3-dB width

**Quick Notes — Fourier Transform** — merged revision notes + key diagrams

**NOTE** Slide numbers refer to Lecture-7-Fourier-Transform.pdf (61 slides, “Lec 7”). The deck covers the FS → FT derivation, the rectangle, time/frequency shifting, modulation and δ(t). Lessons 18–19 add the standard textbook pairs and properties that the TF papers ask (marked “textbook”). Note: CSE 311 papers use f (Hz) and Lathi's sinc(x) = sin x / x; EEE and CSE 219 papers use ω.

# Exam Map — What the TF Papers Ask

The Fourier transform is the heaviest Anik-part topic in CSE 219 (2022-23: Q5(b) 20 + Q6 30 marks; 2023-24: Q1(c) 15 + Q2 30 marks) and appears in every EEE and CSE 311 paper.

| Session (paper) | Questions answered by this note |
|---|---|
| 2023-24 (CSE 219) | Q1(c) FT of rect(t); FT of rect((t−τ)/T)·cos 2πf$_0$t (15) · Q2(a) FT of $x_1 = e^{-at}u(t)$, then of $x_2 = e^{-a\|t\|}$ (10) · Q2(b) Evaluate I(a) = $\int f^2/(a^2+(2\pi f)^2)^2$ df using Parseval (20) |
| 2023-24 (EEE 211) | Q5(a) Quadrature multiplexing (ω_c = 4ω_m): design H(jω); spectra $u_1$, $u_2$, $v_1$, $v_2$, $y_2$ (20) · Q6(a) FT of the periodic impulse signal of Fig. 6(a); amplitude & phase spectra (15) · Q6(b) FT of sin(at)/(πt); energy by Parseval (10) · Q6(c) FT of u(t); prove the integration property (10) |
| 2022-23 (CSE 219) | Q5(b) X(jω)=1 for \|ω\|<3π and $X_1$(jω) of Fig. 1: define $X_1$, inverse FT, $X_1$ via X, $x_1$ via x, $x_1$(t) (20) · Q6(a)–(b) f(t) = $e^{-at}u(t)$: define f(−t); FT of f(t) and f(−t) (15) · Q6(c)–(d) Write sign(t) using f(t), f(−t); FT of sign(t) by properties (15) |
| 2022-23 (EEE 211) | Q2(a) Closed-form FT of x(t) (ramp 0→1, flat to 2) (10) · Q2(b) AM of a down-sampled signal, filter H_f: spectra of x, y, z; energy of z (25) · Q4(a) AM receiver with phase error θ, LPF gain 2: output (15) |
| 2022-23 (CSE 311) | Q1(b) Spectrum of Π(t/2B); practical spectra; how sampling changes the band (18) · Q2(a) Prove g*h ↔ G(f)H(f); LTI output is a convolution (20) · Q2(b) Distortionless vs all-pass system; how to test distortionless (15) |
| 2021-22 (EEE 211) | Q5(a) Derive the FT as the T → ∞ limit of the FS (18) · Q5(b) Inverse FT of the piecewise-constant X(ω) of Fig. 5(b) (17) · Q6(a) Band-pass x_a through c = cos10t, $h_1$, sampler p(t), $h_2$: spectra $x_1$…$x_4$ (20) · Q6(b) h = $e^{2t}u(-t)$: 3-dB bandwidth, 1/e duration, product (15) |
| 2021-22 (CSE 311) | Q1(a) FT of gate of width B and exp. FS of the periodic gate (period 2B); relate (20) · Q2(a) All-pass vs distortionless; frequency response; why ideal filters are unrealizable (20) · Q3(a) Time compression ↔ frequency expansion (quantization part is comms) (20) · Q6(a) Power spectral density; power from PSD (5) |
| 2020-21 (EEE 211) | Q1(a) X(ω) = jω/(ω²−3jω−2): FT of (dx/dt)·cos t (15) · Q1(b) m(t) through $H_1$, $H_2$ (Hilbert pair) × cos/sin: spectrum of y, energy by Parseval (20) · Q2(a) PSD of Σδ(t−nT) (8) · Q2(b) 3-dB bandwidth of $e^{-3t}u(t)$ (7) · Q2(c) Causal y''+6y'+8y = 2x: H(ω); response to $te^{-2t}u(t)$ (20) |
| 2020-21 (CSE 311) | Q3(b) Dirichlet conditions; are they necessary? example (11) · Q4(a) Characteristics of a distortionless system and their physical meaning (20) |
| 2019-20 (EEE 211) | Q6(a) 1/e duration of $Ae^{-t/T}u(t)$, $x_1(3t)$, $x_1(t/2)$ (15) · Q6(b) State and prove the duality property (20) |
| 2019-20 (CSE 311) | Q1(b) FT of cos/sin 2πf$_0$t; Dirichlet; necessity for FT existence (18) · Q2(a) Time delay ↔ linear phase; what if a voice signal lost linear phase? (24) |
| 2018-19 (EEE 211) | Q2(a) Prove E = (1/2π)$\int$\|X\|²dω; energy of sinc(t/2) (10) · Q2(b) x = $e^{-\alpha t}u$ through h = sin2t/(πt): Y(ω); α for half the input energy (10) · Q2(c) SSB generation (ω_f = ω_m): spectrum of y; h_f(t); realizable? (15) · Q4(a) Causal y''+6y'+8y = 2x by FT: h(t); y for x = $te^{-2t}u$ (17) |
| 2018-19 (CSE 311) | Q1(b) Distortionless transmission; is an all-pass system distortionless? (14) · Q4(a) Frequency-shifting property; FDM; dual technique (18) |
| 2017-18 (EEE 211) | Q1(b) Distortionless LTI system; relation between FS coefficients of input and output (8) · Q2(a) Derive Parseval's relation for an aperiodic signal (7) · Q2(b) h = (1/RC)$e^{-t/RC}u(t)$, x = Vu(t): y by frequency-domain method (13) |
| 2017-18 (CSE 311) | Q1(a) Frequency-domain view; spectrum of a pulse vs a pulse train; bandwidth (18) · Q1(b) Triangle h(t) from rectangle g(t) via FT properties (10) · Q1(c) Ideal LPF: transfer function → impulse response (7) |
| 2016-17 (EEE 211) | Q3(b) Distortionless LTI system; FS coefficient relation (10) · Q6(a) FT of rect(t/τ); triangle Δ(t/τ) by the convolution property (12) · Q6(b) Energy of sin(20π(t−1/40))/(π(t−1/40))·cos(30πt) (11) · Q8(a) Impulse response of the ideal band-stop filter (stop 10 < \|ω\| < 20) (12) |
| 2016-17 (CSE 311) | Q1(a) Define bandwidth; sketch two signals of different bandwidth (9) · Q1(b) Real signal: amplitude spectrum even, phase spectrum odd (6) · Q1(c) Duality: FT of sinc²(t) from the triangle pair (6) · Q1(d) Prove an ideal LPF is physically unrealizable (8) · Q1(e) Transfer function of a differentiator (6) |
| 2015-16 (EEE 301) | Q5(a) X = (2+j4ω+ω²)/(3+j4ω−ω²): FT of x*x(t−1), $x(-2t+1)e^{-j2t}$ (10) · Q5(b) Hilbert-pair filters $H_1$, $H_2$ with cos/sin: magnitude spectra $x_1$, $x_2$, $y_1$, $y_2$, y (15) · Q5(c) \|X(ω)\| of sin t · sin(t/2)/(πt²) (10) · Q6(c) y''+6y'+8y = 2x: H(ω); response to $te^{-2t}u$ (13) |
| 2014-15 (EEE 301) | Q5(a) SSB generator: spectrum of y (ω_f = ω_m); h_f(t) (15) · Q5(b) FT of α rect(αt/τ); limits α → ∞ and α → 0 (7) · Q5(c) x = $e^{-\alpha t}u$ into h = sin2t/(πt): FT of output (13) · Q6(a) Modulate → band-pass $h_1$ → demodulate → LPF $h_2$: Y(ω) (16) |

| Session (paper) | Questions answered by this note |
|---|---|
| 2014-15 (CSE 311) | Q5(b) State & prove the time-scaling (dilation) property (10) · Q5(c) FT of $\sin^2(2\pi f_0 t)\cdot g(t)$ (10) · Q5(d) FT of the Dirac delta (5) · Q6(a) Bandwidth of two given spectra (baseband / bandpass) (10) · Q6(b) FT of a periodic signal: derivation steps (i)–(v) (20) · Q7(b) Ideal LPF response to a rectangular pulse; ideal vs real filter (10) · Q1(a) [L-3] How is bandwidth computed? bandwidth of δ(t) (12) · Q1(c) [L-3] $H(f) = (1+k\cos 2\pi fT)e^{-j2\pi ft_d}$: show output is g(t−t_d)+(k/2)[g(t−t_d−T)+g(t−t_d+T)] (13) |
| 2013-14 (EEE 301) | Q7(a) AM system with channel filters: sketch X, Y, Z, V (15) · Q7(b) Show F{sinc(ω_Bt/2π)} = (2π/ω_B) rect(ω/ω_B) (7) · Q7(c) x = $[(1/12)e^{-t/4}+(1/6)e^{-t/2}]u$ into h = $e^{-t/3}u$: Y(ω), y(t) (13) |
| 2013-14 (CSE 311) | Q1(b) Bandwidth calculation; g(t) vs g(at), a>1 (12) · Q1(c)(i) Empirically find the transfer function of an LTI system (8) · Q1(c)(ii) When is an RC circuit approximately distortionless? (10) |
| 2012-13 (EEE 301) | Q5(a) FT of δ(t), 1, $e^{j\omega_0 t}$, cos $\omega_0 t$, sin $\omega_0 t$ (12) · Q5(b) X(ω) of a rectangle; $Y_1$ (doublet), $Y_2$ (triangle) by properties (15) · Q5(c) Convolution in time ↔ multiplication in frequency (8) · Q6(a) Ideal LPF: frequency and impulse response; causality (8) · Q6(c) Hilbert-pair filters: $H_1$, $H_2$, $X_1$, $X_2$, Y (12) |
| 2012-13 (CSE 311) | Q5(b) Convolution in time ↔ multiplication in frequency and vice versa (14) · Q5(c) FT of Π(t/τ) (13) · Q6(b) State & prove time-shifting and frequency-shifting properties (10) · Q7(a) Distortionless: \|H\| constant and linear phase through origin (8) · Q7(c) LPF $(1+k\cos 2\pi fT)e^{-j2\pi ft_d}$: echo output (13) |
| 2011-12 (EEE 301) | Q2(a) g = x cos²t * sin t/(πt), X = 0 for \|ω\|≥1: h mapping x → g (17) · Q4(b) Ideal −90° phase shifter: h(t); output for cos $\omega_0 t$ (17) |
| 2011-12 (CSE 311) | Q5(a) Fourier integral of the rectangular pulse f(t) (8) · Q5(b) FT of g (doublet) from f, then of h (triangle) from g using properties (14) · Q5(c) Which of f, g, h has the lowest bandwidth? (13) · Q6(a) Linearity, time invariance; prove G(ω) = H(ω)F(ω) (18) · Q6(b) Distortionless H(ω); when is an RC circuit distortionless? (17) |
| 2010-11 (EEE 301) | Q2(a) FT of impulse train; energy in Fourier domain; convolution theorem; FT of x cos $\omega_0 t$ (20) |
| 2010-11 (CSE 311) | Q5(a) Bandwidth; which has higher bandwidth, f(t) or f'(t)? (14) · Q5(b) FT of g(t) from the triangle pair via shift and scaling; inverse time–bandwidth (21) · Q6(a) Impulse response of ideal LPF with linear phase; why unrealizable (15) · Q6(b) g(t) of Fig. 5(b) through H(ω) = jω: output (8) |

**EXAM FOCUS** Learn first: (1) the pair table and the properties in Lesson 18 — CSE 219 asks “use the properties, name them” (sign(t) from $e^{-at}u$, $e^{-a|t|}$ from $e^{-at}u$, Parseval integrals) · (2) modulation/frequency shift (2023-24 Q1(c), EEE spectra-sketch questions every year) · (3) inverse FT of a given spectrum (2022-23 Q5(b)) · (4) distortionless systems, ideal LPF unrealizable, bandwidth (CSE 311 every year).

# Lesson 16 — From Fourier Series to Fourier Transform

*Slides 2–18 (Lec 7) · existence and periodic signals: textbook*

Fourier series need x(t + T) = x(t); real signals are often one-time events (sl. 5). Trick (sl. 6–17): repeat one pulse with period T, take the FS, let T → ∞.

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a_k = (1/T)∫_{-T/2}^{T/2} x(t)e^{-jkω₀t}dt = (1/T)X(jkω₀),   X(jω) ≜ ∫ x(t)e^{-jωt}dt
⇒ T·a_k are samples of a smooth envelope X(jω) at ω = kω₀ (sl. 12)
x̃_T(t) = Σ a_k e^{jkω₀t} = (1/2π) Σ X(jkω₀) e^{jkω₀t} Δω,   Δω = ω₀ = 2π/T
T → ∞: Δω → dω, kω₀ → ω, Σ → ∫  ⇒  x(t) = (1/2π)∫ X(jω)e^{jωt} dω
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$$\mathbf{X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt} \qquad \mathbf{x(t) = (1/2\pi)\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega}$$

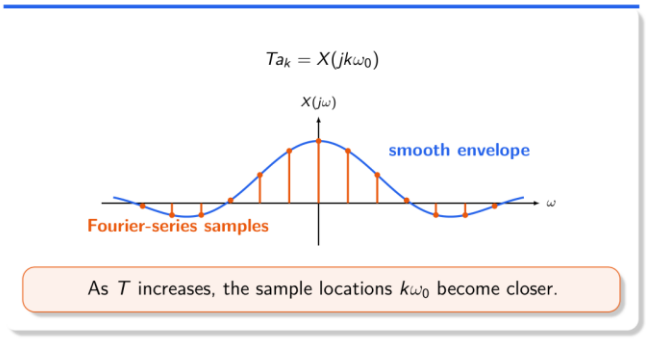



*Figure 16.1 — Fourier-series stems are samples of the smooth envelope X(jω) (Lec 7, sl. 12)*

Why raw coefficients shrink (sl. 13–14): $a_k = (\Delta\omega/2\pi)X(jk\omega_0)$ — each stem is the area of a bin of width Δω, so X(jω) is a **spectral density**. As T grows the lines crowd together and shorten.

## Existence and Periodic Signals (textbook)

- Dirichlet conditions for the FT: $\int|x|dt < \infty$, finitely many extrema and finite jumps in any finite interval. Sufficient only: cos $\omega_0 t$, u(t), δ-trains have transforms containing δ(ω).
- Periodic x with FS $a_k$: $X(j\omega) = 2\pi \Sigma a_k \delta(\omega - k\omega_0)$ (each harmonic $e^{jk\omega_0 t} \leftrightarrow 2\pi\delta(\omega - k\omega_0)$).
- Impulse train $\Sigma\delta(t - nT) \leftrightarrow (2\pi/T)\Sigma\delta(\omega - 2\pi k/T)$: a train stays a train.

## Exam Approach

1. Derivation question: write $a_k$ for the periodic extension, identify $T\cdot a_k = X(jk\omega_0)$, substitute $1/T = \Delta\omega/2\pi$ into the synthesis sum, take the limit.
2. Periodic signal: find $a_k$ first, then put $2\pi a_k$ impulses at $k\omega_0$.
3. State Dirichlet conditions and whether they are necessary.

## Practice — Lesson 16

*(a) Actual — EEE 211 2021-22 Q5(a). (b) Actual — EEE 211 2023-24 Q6(a). (c) Actual — CSE 311 2021-22 Q1(a), 2017-18 Q1(a). (d) Actual — CSE 311 2020-21 Q3(b), 2019-20 Q1(b), 2014-15 Q6(b).*

**(a)** What happens to $\tilde{x}(t) = \Sigma C_n e^{jn\omega_0 t}$ if the waveform stays the same but T grows? Show that as T → ∞ one period becomes $x(t) = (1/2\pi)\int X(\omega)e^{j\omega t}d\omega$. (18)

**(b)** x(t) is periodic with impulses +1 at t = 1, 2 and −1 at t = 3, 4 (period 4). Find its FT; plot amplitude and phase spectra. (15)

**(c)** FT of a gate of width B and the exponential FS of the gate repeated with period 2B; relate them. Spectrum of a pulse vs a pulse train. (20)

**(d)** Dirichlet conditions — necessary? Represent cos $2\pi f_0 t$ by an FT. FT of a periodic signal.

## ✓ Solution

**(a)** As T increases the lines $k\omega_0 = 2\pi k/T$ crowd together and each $C_n = (1/T)X(jn\omega_0)$ shrinks, but $T \cdot C_n$ stays on the envelope $X(j\omega)$. Derivation in the grey box above: substitute $C_n = (\Delta\omega/2\pi)X(jn\omega_0)$ and let $\Delta\omega \to 0$.

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(b) T = 4, ω₀ = π/2. One period: δ(t−1) + δ(t−2) − δ(t−3) − δ(t−4)
    a_k = ¼[e^{−jkπ/2} + e^{−jkπ} − e^{−j3kπ/2} − 1]
    even k: a_k = 0;   odd k: a_k = ½(−j)^k − ½
    a±1 = −½ ∓ j½ (|a| = 0.707, ∠ = ∓135°);   a±3 = −½ ± j½ (∠ = ±135°);  pattern repeats
every 4
    X(jω) = 2π Σ_{odd k} a_k δ(ω − kπ/2): impulses of area 2π/√2 = 4.44 at ω = ±π/2, ±3π/2, …
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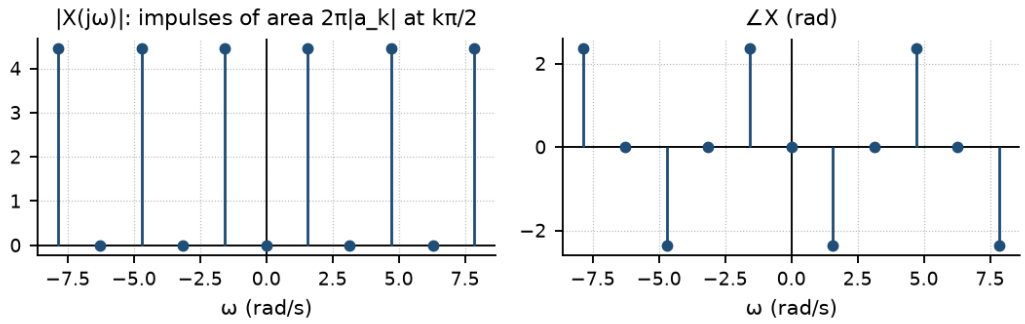



*Figure 16.2 — EEE 211 2023-24 Q6(a): amplitude and phase spectra*

**(c)** Gate of width B: $X(f) = B \cdot \sin(\pi fB)/(\pi fB)$ (continuous). Repeated with period $T_0 = 2B$: $D_n = (1/T_0)X(n/T_0) = \frac{1}{2} \cdot \sin(n\pi/2)/(n\pi/2)$ — lines at $n/(2B)$ Hz whose heights are samples of $X(f)/T_0$. A single pulse has a continuous spectrum; the train has a line spectrum under the same sinc envelope. Bandwidth (first null) = 1/B Hz in both.

**(d)** Dirichlet conditions as above — sufficient, not necessary. $\cos 2\pi f_0 t$ is not absolutely integrable, yet $\cos 2\pi f_0 t \leftrightarrow \frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$ ($\sin \leftrightarrow (1/2j)[\delta(f - f_0) - \delta(f + f_0)]$). Periodic signals: $X = 2\pi\Sigma a_k \delta(\omega - k\omega_0)$.

# Lesson 17 — Rectangles, Shifts and Modulation

*Slides 19–44 (Lec 7)*

## 1. Frequency Shift and Modulation (sl. 19–28)

$$x(t)e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$$
$$x(t)\cos \omega_0 t \leftrightarrow \tfrac{1}{2}X(j(\omega - \omega_0)) + \tfrac{1}{2}X(j(\omega + \omega_0))$$

Each component $e^{j\omega t}$ times $e^{j\omega_0 t}$ becomes $e^{j(\omega+\omega_0)t}$: the whole spectrum slides right by $\omega_0$. A cosine is two rotating exponentials, so it creates two half-height copies at $\pm\omega_0$ (damped pendulum, ringing circuits, AM radio).

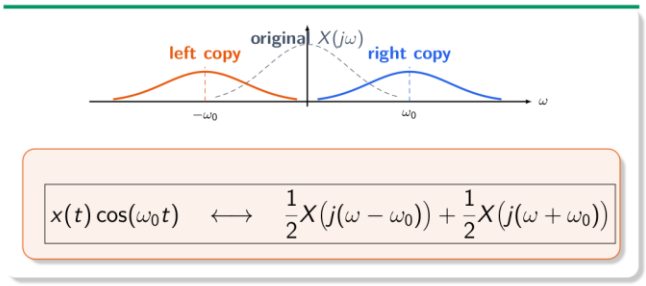



*Figure 17.1 — Cosine modulation creates two spectral copies (Lec 7, sl. 27)*

## 2. The Rectangle (sl. 29–38)

$$x(t) = 1 \text{ for } |t| < T/2 \;\leftrightarrow\; X(j\omega) = 2\sin(\omega T/2)/\omega \;\; (= T \text{ at } \omega = 0)$$

Wider rectangle → narrower main lobe (first zero at $2\pi/T$). Widths 2 and 6: $2\sin \omega/\omega$ and $2\sin 3\omega/\omega$. Shifted rectangle $4 < t < 10$ = width-6 rectangle shifted by 7: $X = e^{-j7\omega}\cdot 2\sin 3\omega/\omega$, $X(0) = 6$.

$$\text{Time shift: } x(t - t_0) \leftrightarrow e^{-j\omega t_0}X(j\omega) \quad (|X| \text{ unchanged, phase } -\omega t_0\text{: linear in } \omega)$$

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Lego block (sl. 36–38): height-1 block on (1,4) + height-0.5 block on (2,3), both centred at
2.5
X(jω) = e^{-j2.5ω}[2 sin(1.5ω)/ω + sin(0.5ω)/ω]
Windowed cosine r(t)cos ω₀t (sl. 39–40): ½[R(j(ω − ω₀)) + R(j(ω + ω₀))] — two sincs centred at
±ω₀
Impulse (sl. 41–44): δ(t) ↔ 1 (flat — a spike needs all frequencies);  δ(t − t₀) ↔ e^{-jωt₀}
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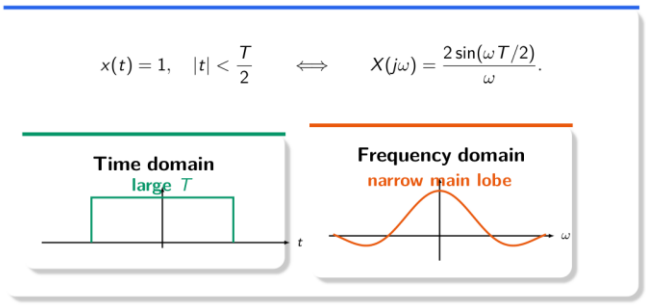



*Figure 17.2 — Wider rectangle, narrower spectrum (Lec 7, sl. 31)*

**MEMORIZE** Proof of time shift: $\int x(t-t_0)e^{-j\omega t}dt$, put $\tau = t - t_0 \Rightarrow e^{-j\omega t_0}\int x(\tau)e^{-j\omega\tau}d\tau$. Proof of frequency shift: $\int x\, e^{j\omega_0 t}e^{-j\omega t}dt = \int x\, e^{-j(\omega-\omega_0)t}dt = X(j(\omega-\omega_0))$. In Hz: $\Pi(t/\tau) \leftrightarrow \tau\,\mathrm{sinc}(f\tau)$ with $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$; $g(t-t_0) \leftrightarrow G(f)e^{-j2\pi f t_0}$.

## Exam Approach

1. Decompose the signal into rectangles/triangles/known pulses.
2. Transform each centred piece, then multiply by $e^{-j\omega t_c}$ for its centre.
3. Modulation: write ½ copies at $\pm\omega_0$; sketch both copies with half height.
4. Name every property used.

## Practice — Lesson 17

*(a) Actual — CSE 219 2023-24 Q1(c). (b) Actual — CSE 311 2019-20 Q2(a), 2012-13 Q6(b). (c) Actual — EEE 211 2023-24 Q5(a). (d) Actual — EEE 211 2022-23 Q2(b), Q4(a). (e) Actual — EEE 211 2021-22 Q6(a). (f) Practice set 7.1–7.4; CSE 311 2014-15 Q5(c), 2012-13 Q5(c), 2018-19 Q4(a), 2011-12 Q5(a), 2022-23 Q1(b).*

**(a)** FT of $x_1 = \mathrm{rect}(t)$; then $x_2 = \mathrm{rect}((t-\tau)/T)\cdot\cos 2\pi f_0 t$ by properties. (15)

**(b)** Show that a time delay causes a linear phase shift. What would happen to a voice signal if this did not hold? (24)

**(c)** Quadrature multiplexing: $u_1 = x_1\cos \omega_ct$, $u_2 = x_2 \sin \omega_ct$, $r = u_1 + u_2$; receiver multiplies by cos ω_ct and sin ω_ct and filters with $H(j\omega)$. $X_1$ = rect of height 1 on $|\omega| < \omega_m$, $X_2$ = triangle peak 1, $\omega_c = 4\omega_m$. (i) Design H so that $y_1 = x_1$, $y_2 = x_2$. (ii) Spectra of $u_1, u_2, v_1, v_2, y_2$. (20)

**(d)** (i) m(t) (triangle spectrum, height A, edge ω_m) is down-sampled by 2 ($x(t) = m(2t)$), multiplied by cos $\omega_0 t$ and filtered by H_f = B on $\omega_0 < |\omega| < \omega_0 + \omega_f$, with $\omega_m = \omega_f = \omega_0/2$. Spectra of x, y, z; energy of z. (25) (ii) AM: x cos ω_ct, receiver oscillator cos(ω_ct − θ), LPF gain 2: output? (15)

**(e)** x_a: triangles of height 1 centred at ±10 (width ±2). $x_1$ = x_a cos 10t → $h_1 = \sin 2t/(\pi t)$ → $x_2$ → × $p(t) = \Sigma\delta(t - 2\pi n/5)$ → $x_3$ → $h_2 = 2\sin(2t)\cos(10t)/(\pi t)$ → $x_4$. Sketch the spectra. (20)

**(f)** Shifted rect, Lego block, r(t)cos 3t, impulses; $\sin^2(2\pi f_0 t)g(t)$; $\Pi(t/\tau)$.

### ✓ Solution

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(a) rect(t) ↔ sin(πf)/(πf) = sinc f   (2 sin(ω/2)/ω in ω)
    time scaling: rect(t/T) ↔ T sinc(fT);  time shift by τ: ↔ T sinc(fT)e^{-j2πfτ}
    modulation by cos 2πf₀t:
    X₂(f) = (T/2)[sinc((f − f₀)T)e^{-j2π(f-f₀)τ} + sinc((f + f₀)T)e^{-j2π(f+f₀)τ}]
```

**(b)** Proof: as in the MEMORIZE box, $x(t-t_0) \leftrightarrow X(\omega)e^{-j\omega t_0}$: magnitude unchanged, phase $-\omega t_0$ — a straight line through the origin with slope $-t_0$. Meaning: every component is delayed by the same $t_0$ (group delay $-d\varphi/d\omega = t_0$), so the waveform keeps its shape. A voice signal is a sum of many frequencies; if the phase were not linear, different frequencies would arrive with different delays (dispersion): syllables smear, transients spread, speech sounds distorted even though every amplitude is intact.

```
(c) v₁ = r cos ω_ct = x₁cos²ω_ct + x₂ sin ω_ct cos ω_ct = ½x₁ + ½x₁cos 2ω_ct + ½x₂ sin 2ω_ct
    v₂ = r sin ω_ct = ½x₂ − ½x₂cos 2ω_ct + ½x₁ sin 2ω_ct
(i) H(jω) = 2 for |ω| < ω_f, 0 otherwise, with ω_m ≤ ω_f ≤ 2ω_c − ω_m = 7ω_m (ideal LPF, gain
2)
(ii) |U₁| = ½ rect around ±4ω_m (width ±ω_m); |U₂| = ½ triangles at ±4ω_m;
     |V₁|: ½X₁ at baseband + copies of height ¼ (x₁ and x₂ parts) around ±8ω_m; same for V₂
with x₂ at baseband;
     |Y₂| = triangle of height 1 on |ω| < ω_m  (= X₂)
```

(d)(i) X(ω) = ½M(ω/2): triangle of height A/2 reaching ±2ω_m = ±$\omega_o$
Y(ω) = ½X(ω − $\omega_o$) + ½X(ω + $\omega_o$): triangles of height A/4 centred at ±$\omega_o$ spanning 0…2$\omega_o$
Z(ω) = B·Y on $\omega_o$ < |ω| < 1.5$\omega_o$: the upper half-triangle from A/4 (at $\omega_o$) down to A/8 (at 1.5$\omega_o$) — an SSB-like signal
E_z = (1/2π)·2∫_{$\omega_o$}^{1.5$\omega_o$} (AB/4)²(1 − (ω − $\omega_o$)/$\omega_o$)² dω = ($A^2B^2\omega_o$/16π)(7/24) = $7A^2B^2\omega_o$/(384π)
(ii) x cos ω_ct · cos(ω_ct − θ) = ½x[cos θ + cos(2ω_ct − θ)]; LPF (gain 2) removes the 2ω_c term:
y(t) = x(t) cos θ  (θ = 90° gives zero output — the quadrature null)

(e) $X_1$ = ½X_a(ω − 10) + ½X_a(ω + 10): a triangle of height 1 at 0 (two halves add) + height-½ triangles at ±20
$h_1$ = sin 2t/(πt) ↔ ideal LPF, gain 1, |ω| < 2 ⇒ $X_2$ = triangle of height 1 on |ω| < 2
p(t): period 2π/5 ⇒ P = 5Σδ(ω − 5k);  $X_3$ = (1/2π)$X_2$ * P = (5/2π)Σ $X_2$(ω − 5k)  (copies every 5, height 0.796,
no overlap since 5 > 4)
$h_2$ = 2 sin 2t cos 10t/(πt) ↔ gain 1 on 8 < |ω| < 12 ⇒ keeps the copies at ±10:
$X_4$ = (5/2π)[triangles at ±10] ⇒ $x_4$(t) = (5/2π)x_a(t) ≈ 0.796 x_a(t)  (the band-pass signal is recovered)

(f) Practice set: x = 1 on (4,10): e^{−j7ω}2 sin3ω/ω, X(0) = 6;  Lego with heights 3, 4.5:
e^{−j2.5ω}[6 sin1.5ω/ω + 3 sin0.5ω/ω];  r(t)cos 3t: sin(ω − 3)/(ω − 3) + sin(ω + 3)/(ω + 3);
3δ(t+2) − 2δ(t−1) ↔ 3e^{j2ω} − 2e^{−jω}, X(0) = 1
$\sin^2(2\pi f_o t)g(t)$ = ½g − ½g cos $4\pi f_o t$ ↔ ½G(f) − ¼[G(f − $2f_o$) + G(f + $2f_o$)]
Π(t/τ) ↔ τ sinc(πfτ) (Lathi) = τ sin(πfτ)/(πfτ);  Π(t/2B) ↔ 2B sin(2πBf)/(2πBf)

# Lesson 18 — Transform Pairs and Properties Toolbox

*Textbook (Oppenheim & Willsky Tables 4.1–4.2), built on Lec 7*

| x(t) | X(jω) | x(t) | X(jω) |
|---|---|---|---|
| $\delta(t)$ | $1$ | $1$ | $2\pi\delta(\omega)$ |
| $e^{-at}u(t)$, $a > 0$ | $1/(a + j\omega)$ | $e^{at}u(-t)$ | $1/(a - j\omega)$ |
| $e^{-a\lvert t\rvert}$ | $2a/(a^2 + \omega^2)$ | $te^{-at}u(t)$ | $1/(a + j\omega)^2$ |
| $\mathrm{sgn}(t)$ | $2/(j\omega)$ | $u(t)$ | $\pi\delta(\omega) + 1/(j\omega)$ |
| rect: 1 for $\lvert t\rvert < T/2$ | $2\sin(\omega T/2)/\omega$ | $\sin(Wt)/(\pi t)$ | 1 for $\lvert\omega\rvert < W$ |
| triangle $1 - \lvert t\rvert/\tau$, $\lvert t\rvert < \tau$ | $\tau\cdot[\sin(\omega\tau/2)/(\omega\tau/2)]^2$ | $e^{j\omega_0 t}$ | $2\pi\delta(\omega - \omega_0)$ |
| $\cos\omega_0 t$ | $\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$ | $\sin\omega_0 t$ | $(\pi/j)[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$ |
| $\Sigma\delta(t - nT)$ | $(2\pi/T)\Sigma\delta(\omega - 2\pi k/T)$ | $e^{-at^2}$ | $\sqrt{(\pi/a)}\, e^{-\omega^2/4a}$ |

| Property | x(t) | X(jω) |
|---|---|---|
| Linearity | $ax + by$ | $aX + bY$ |
| Time shift / frequency shift | $x(t - t_0)$ / $x\, e^{j\omega_0 t}$ | $e^{-j\omega t_0}X$ / $X(j(\omega - \omega_0))$ |
| Scaling | $x(at)$ | $(1/\lvert a\rvert)X(j\omega/a)$ (compress in time ⇔ stretch in frequency) |
| Reversal | $x(-t)$ | $X(-j\omega)$ (= X* for real x) |
| Duality | $X(jt)$ | $2\pi x(-\omega)$ |
| Differentiation | $dx/dt$ | $j\omega X$ |
| Frequency differentiation | $-jt\, x(t)$ | $dX/d\omega$ |
| Integration | $\int_{-\infty}^{t} x\, d\tau$ | $X/(j\omega) + \pi X(0)\delta(\omega)$ |
| Convolution | $x * h$ | $XH$ |
| Multiplication | $x\cdot y$ | $(1/2\pi)X * Y$ |
| Parseval | $\int\lvert x\rvert^2 dt$ | $(1/2\pi)\int\lvert X\rvert^2 d\omega$ (= $\int\lvert X(f)\rvert^2 df$) |
| Real signal | x real | $\lvert X\rvert$ even, $\angle X$ odd; x even ⇒ X real |

**MEMORIZE** Proofs to know: (i) duality — from $x(t) = (1/2\pi)\int X(j\omega)e^{j\omega t}d\omega$, $2\pi x(-t) = \int X(j\omega)e^{-j\omega t}d\omega$; rename $t \leftrightarrow \omega$: $2\pi x(-\omega) = \int X(jt)e^{-j\omega t}dt = F\{X(jt)\}$. (ii) convolution — swap the order of integration and use the time shift. (iii) Parseval — $\int x\, x^*\, dt$ with $x^*$ written as the conjugate inverse transform. (iv) scaling — substitute $\tau = at$ ($\lvert a\rvert$ because $a < 0$ flips the limits). (v) real x: $X(-j\omega) = \int x\, e^{j\omega t} = X^*(j\omega)$.

## Exam Approach

1. Recognise the signal as a known pair ± shift/scale/modulate/differentiate.
2. Write each step with its property name (CSE 219 gives marks for naming).
3. For a ratio $X(\omega) = N/D$: partial fractions in $j\omega$, invert term by term.
4. Parseval integrals: identify $\int\lvert X\rvert^2$ as the energy of a known x(t) and integrate in time instead.

## Practice — Lesson 18

*(a) Actual — CSE 219 2022-23 Q6. (b) Actual — CSE 219 2023-24 Q2. (c) Actual — CSE 219 2022-23 Q5(b). (d) Actual — EEE 211 2023-24 Q6(b)–(c), 2022-23 Q2(a), 2021-22 Q5(b), 2020-21 Q1(a), 2019-20 Q6(b).*

**(a)** $f(t) = e^{-at}u(t)$, $a > 0$; $\mathrm{sign}(t) = -1$ ($t < 0$), $1$ ($t > 0$). (a) Define $f(-t)$. (b) FT of f and $f(-t)$. (c) sign(t) in terms of $f(t)$, $f(-t)$. (d) FT of sign(t) by properties (name them). (30)

**(b)** $x_1 = e^{-at}u(t)$, $x_2 = e^{-a\lvert t\rvert}$. (a) FT of $x_1$, then of $x_2$ (name the properties). (b) Evaluate $I(a) = \int f^2/(a^2 + (2\pi f)^2)^2\, df$ using Parseval. (30)

**(c)** X(jω) = 1 for |ω| < 3π; $|X_1(j\omega)|$ rises linearly from 0 at ω = 0 to 9π at |ω| = 3π (zero outside), $\angle X_1$ = π/2 (ω > 0), −π/2 (ω < 0). (i) Define $X_1$. (ii) x(t). (iii) $X_1$ via X. (iv) $x_1$ via x. (v) $x_1(t)$. (20)

**(d)** FT of sin(at)/(πt) and its energy; FT of u(t) and the integration property; closed-form FT of a ramp-and-hold pulse (0 → 1 on [0,1], 1 on [1,2]); inverse FT of X = 1 on 3 < |ω| < 7 plus 1 more on 4 < |ω| < 6; X = jω/(ω² − 3jω − 2): FT of x′(t)cos t; duality.

## ✓ Solution

```
(a)(a) f(−t) = e^{at} for t < 0, 0 for t > 0  (= e^{at}u(−t))
   (b) F{f} = ∫₀^∞ e^{−at}e^{−jωt}dt = 1/(a + jω);   time reversal: F{f(−t)} = F(−jω) = 1/(a − jω)
   (c) sign(t) = lim_{a→0} [f(t) − f(−t)]
   (d) linearity + time reversal: F{f(t) − f(−t)} = 1/(a + jω) − 1/(a − jω) = −2jω/(a² + ω²)
       a → 0: F{sign(t)} = −2jω/ω² = 2/(jω)   (X(0) = 0: sign has zero average)
(b)(a) X₁ = 1/(a + jω); x₂ = x₁(t) + x₁(−t) (linearity + reversal) ⇒ X₂ = 1/(a+jω) + 1/(a−jω) = 2a/(a² + ω²)
   (b) Differentiation: dx₂/dt ↔ jω·2a/(a² + ω²) = j2πf·2a/(a² + (2πf)²)
       |F{dx₂/dt}|² = 16π²a² f²/(a² + (2πf)²)²
       Parseval (in f): ∫|dx₂/dt|²dt = ∫|F|²df.  dx₂/dt = −a sgn(t)e^{−a|t|} ⇒ ∫a²e^{−2a|t|}dt = a
       ⇒ 16π²a²·I(a) = a  ⇒  I(a) = 1/(16π²a)
```

```
(c)(i)  X₁(jω) = 3|ω|e^{jπ sgn ω/2} = 3jω for |ω| < 3π, 0 otherwise
   (ii) x(t) = (1/2π)∫_{−3π}^{3π} e^{jωt}dω = sin(3πt)/(πt)
   (iii) X₁(jω) = 3jω X(jω)
   (iv) differentiation property: x₁(t) = 3 dx/dt
   (v)  x₁(t) = 3 d/dt[sin 3πt/(πt)] = 3(3πt cos 3πt − sin 3πt)/(πt²)
```

```
(d) sin(at)/(πt) ↔ 1 for |ω| < a  (duality of the rectangle);  E = (1/2π)·2a = a/π
    u(t) = ½ + ½sgn(t) ↔ πδ(ω) + 1/(jω);  ∫_{−∞}^{t}x = x * u ↔ X(jω)[πδ(ω) + 1/(jω)] = X/(jω) + πX(0)δ(ω)
    ramp-and-hold: x′ = u(t) − u(t−1) − δ(t−2) ⇒ jωX = (1 − e^{−jω})/(jω) − e^{−j2ω}
        X(ω) = (e^{−jω} − 1)/ω² + j e^{−j2ω}/ω
    band rects: a rect on a < |ω| < b ↔ [sin bt − sin at]/(πt) ⇒ x(t) = [sin 7t − sin 3t + sin 6t − sin 4t]/(πt)
    x′(t)cos t: x′ ↔ G = jωX = −ω²/(ω² − 3jω − 2) ⇒ F{x′cos t} = ½G(ω − 1) + ½G(ω + 1)
    duality: proof in the MEMORIZE box (EEE 2019-20 Q6(b), 20 marks — add the example rect ↔ sinc ⇒ sinc ↔ 2π rect)
```

```
Older quick answers: −90° phase shifter H = −j sgn ω ⇒ h = 1/(πt), cos ω₀t → sin ω₀t;
  sin t·sin(t/2)/(πt²) = π·[sin t/(πt)]·[sin(t/2)/(πt)] ↔ ½·(rect|ω|<1 * rect|ω|<½): |X| = ½ for |ω| < ½,
  falling linearly to 0 at |ω| = 3/2;  α rect(αt/τ) ↔ τ·sin(ωτ/2α)/(ωτ/2α): α → ∞ gives τ (flat; τδ(t)),
  α → 0 shrinks the spectrum onto ω = 0;  energy of sin(20π(t − 1/40))/(π(t − 1/40))·cos 30πt = 10
  (rect of height 1 on |ω| < 20π, shifted to ±30π with height ½: (1/2π)·2·¼·40π)
```

# Lesson 19 — LTI Systems in Frequency: Filters, Distortionless, Bandwidth

*Slides 45–59 (Lec 7: applications) · frequency response, filters, distortionless, PSD: textbook*

Applications on the slides: audio and image denoising (remove noise frequencies), JPEG (drop small high-frequency coefficients), MRI (data measured in k-space = frequency domain), music fingerprinting (peak patterns in a spectrogram), vibration diagnosis (faults change the spectrum), ECG (notch out 50 Hz mains hum).

$$y = x * h \iff Y(j\omega) = H(j\omega)X(j\omega) \qquad e^{j\omega t} \to H(j\omega)e^{j\omega t}$$

## Distortionless Transmission (textbook)

Distortionless: $y(t) = K\,x(t - t_d) \Rightarrow H(j\omega) = K\,e^{-j\omega t_d}$: $|H| = K$ constant and phase $-\omega t_d$ linear through the origin (constant group delay $t_d$). An **all-pass** system has $|H|$ constant but its phase need not be linear, so it is distortionless only if its phase is linear — otherwise it causes phase (delay) distortion.

RC low-pass $H = 1/(1 + j\omega RC)$: for $\omega RC \ll 1$, $|H| \approx 1$ and $\angle H \approx -\omega RC$ — approximately distortionless (delay RC) for signals whose bandwidth $B \ll 1/(2\pi RC)$.

## Ideal Filters (textbook)

```
Ideal LPF H = 1 (|ω| < ω_c), linear phase e^{-jωt_d}:  h(t) = sin(ω_c(t − t_d))/(π(t − t_d))
  h ≠ 0 for t < 0 for every finite t_d ⇒ non-causal ⇒ physically unrealizable (Paley-Wiener:
|H| cannot be
  zero over a band). Practical filters approximate it with a large delay and a truncated sinc.
Ideal band-stop (stop 10 < |ω| < 20): h = δ(t) − [sin 20t − sin 10t]/(πt)
Echo filter H = (1 + k cos 2πfT)e^{-j2πft_d} = e^{-j2πft_d}[1 + (k/2)(e^{j2πfT} + e^{-j2πfT})]
  ⇒ y = g(t − t_d) + (k/2)[g(t − t_d + T) + g(t − t_d − T)]  (a band-limited input inside |f|
< B)
```

## Bandwidth, Duration, PSD (textbook)

- Bandwidth = range of frequencies holding (most of) the energy: first-null, 3-dB ($|X|^2$ falls to half) or essential bandwidth. $\delta(t)$ has infinite bandwidth (flat spectrum).
- Scaling: $x(at)$, $a > 1$, is shorter ⇒ wider spectrum (time–bandwidth product constant). A derivative boosts high frequencies ($|j\omega X|$) ⇒ larger bandwidth. Smoother signals (rect → triangle) have narrower bandwidth.
- PSD of a power signal: $S(\omega) = \lim |X_T(\omega)|^2/T$; $P = (1/2\pi)\int S\,d\omega$. Periodic x: $S = 2\pi\Sigma|a_k|^2\delta(\omega - k\omega_0)$.
- Empirical H: drive with sinusoids $e^{j\omega t}$ (or an impulse/white noise), measure output amplitude ratio and phase shift at each ω.

## Practice — Lesson 19

*(a) Actual — CSE 311 2022-23 Q2(b), 2021-22 Q2(a), 2020-21 Q4(a), 2018-19 Q1(b), 2012-13 Q7(a). (b) Actual — CSE 311 2016-17 Q1(d), 2017-18 Q1(c), 2010-11 Q6(a). (c) Actual — EEE 211 2020-21 Q2(c). (d) Actual — EEE 211 2021-22 Q6(b), 2020-21 Q2(a)–(b), 2019-20 Q6(a); CSE 311 2021-22 Q3(a), 6(a). (e) Older: echo filter, bandwidth questions, RC, band-stop, $e^{-at}u$ through sin2t/(πt).*

**(a)** Characterize a distortionless system; is an all-pass system distortionless? Physical interpretation.

**(b)** Ideal LPF: transfer function → impulse response; prove it is unrealizable.

**(c)** Causal $y'' + 6y' + 8y = 2x$: $H(\omega)$; response to $x = te^{-2t}u(t)$. (20)

**(d)** $h = e^{2t}u(-t)$: 3-dB bandwidth, 1/e duration, product; PSD of $\Sigma\delta(t - nT)$; 3-dB bandwidth of $e^{-3t}u(t)$; 1/e durations of $Ae^{-t/T}u(t)$, $x_1(3t)$, $x_1(t/2)$; time compression ↔ frequency expansion.

**(e)** Older (see solutions).

## ✓ Solution

**(a)** $y = Kx(t - t_d)$ for every input ⇒ $Y = Ke^{-j\omega t_d}X$ ⇒ $H = Ke^{-j\omega t_d}$: constant gain (no amplitude distortion) and phase linear in ω through the origin (every component delayed by the same $t_d$, no phase distortion). An all-pass filter only guarantees constant $|H|$; e.g. $H = (a - j\omega)/(a + j\omega)$ has phase $-2\tan^{-1}(\omega/a)$, not linear ⇒ frequencies are delayed differently ⇒ **not distortionless** (unless the phase is linear over the signal band, where it then behaves as a pure delay).

**(b)** $H = e^{-j\omega t_d}$ for $|\omega| < \omega_c$ ⇒ $h(t) = (1/2\pi)\int_{-\omega_c}^{\omega_c} e^{j\omega(t-t_d)}d\omega = \sin \omega_c(t - t_d)/(\pi(t - t_d))$. The sinc extends to $t \to -\infty$, so $h(t) \neq 0$ for $t < 0$: the filter responds before the impulse arrives ⇒ non-causal ⇒ not realizable. A pulse input gives an output with ripples (Gibbs) and a non-zero response before the pulse.

```
(c) H(ω) = 2/((jω)² + 6jω + 8) = 2/((jω + 2)(jω + 4));  X = 1/(jω + 2)²
    Y = 2/((jω + 2)³(jω + 4)) = −¼/(jω+4) + ¼/(jω+2) − ½/(jω+2)² + 1/(jω+2)³
    y(t) = [−¼e^{−4t} + ¼e^{−2t} − ½te^{−2t} + ½t²e^{−2t}] u(t)
(d) H = 1/(2 − jω): |H|² = 1/(4 + ω²) halves at ω = 2 ⇒ 3-dB bandwidth 2 rad/s; h(0) = 1 falls
to 1/e at
    t = −½ ⇒ duration ½; product = 1 (fixed for this shape: the time–bandwidth relation)
    PSD of Σδ(t − nT): S(ω) = (2π/T²)Σδ(ω − 2πk/T)  (power 1/T² in each line)
    e^{−3t}u: |X|² = 1/(9 + ω²) ⇒ 3 rad/s (0.477 Hz);  durations: T, T/3, 2T (compress ×3 ⇒
duration ÷3,
    bandwidth ×3);  g(at) ↔ (1/|a|)G(f/a): compression in time = expansion in frequency
```

```
(e) x = e^{−αt}u through sin 2t/(πt) (ideal LPF |ω| < 2): Y = 1/(α + jω), |ω| < 2;
    E_y = (1/π)(1/α)tan⁻¹(2/α); half of E_x = 1/(2α) ⇒ tan⁻¹(2/α) = π/4 ⇒ α = 2
    x = [(1/12)e^{−t/4} + (1/6)e^{−t/2}]u, h = e^{−t/3}u ⇒ y = (e^{−t/4} − e^{−t/2})u(t)
    RC, x = Vu: y = V(1 − e^{−t/RC})u(t);  band-stop: h = δ(t) − [sin20t − sin10t]/(πt)
    bandwidth of δ: ∞;  f vs f′: f′ has larger bandwidth (|ωF| grows);  g vs g(at), a > 1:
g(at) wider
```

# Quick Notes — Fourier Transform

*Revision points for Lessons 16–19.*

## L16 · FS → FT

- $T \cdot a_k = X(jk\omega_0)$; $X = \int x\, e^{-j\omega t}$; $x = (1/2\pi)\int X\, e^{j\omega t}$.
- Periodic: $X = 2\pi \Sigma a_k \delta(\omega - k\omega_0)$. Dirichlet sufficient, not necessary.

## L17 · Shifts

- $x(t - t_0) \leftrightarrow e^{-j\omega t_0} X$; $x\, e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0)$; $x \cos \omega_0 t \leftrightarrow \tfrac{1}{2}[X(\omega - \omega_0) + X(\omega + \omega_0)]$.
- rect width $T \leftrightarrow 2\sin(\omega T/2)/\omega$; $\delta \leftrightarrow 1$; (4,10) rect $\leftrightarrow e^{-j7\omega} 2 \sin 3\omega/\omega$.

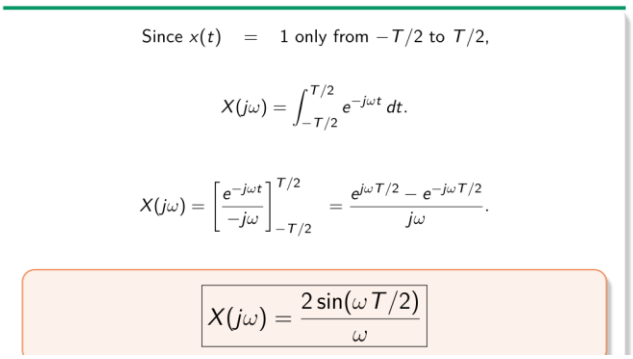
## L18 · Toolbox

- $e^{-at}u \leftrightarrow 1/(a + j\omega)$; $e^{-a|t|} \leftrightarrow 2a/(a^2 + \omega^2)$; sgn $\leftrightarrow 2/(j\omega)$; $u \leftrightarrow \pi\delta + 1/(j\omega)$.
- Duality $X(jt) \leftrightarrow 2\pi x(-\omega)$; $d/dt \leftrightarrow j\omega$; $\int \leftrightarrow X/(j\omega) + \pi X(0)\delta$; Parseval $E = (1/2\pi)\int |X|^2$.
- CSE 219 2023-24 Q2(b): $I(a) = 1/(16\pi^2 a)$.

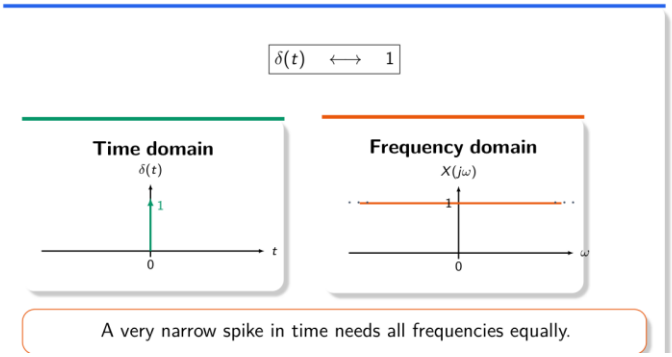
## L19 · Filters

- Distortionless $H = Ke^{-j\omega t_d}$; all-pass ≠ distortionless unless phase linear.
- Ideal LPF $h = \sin \omega_c(t - t_d)/(\pi(t - t_d))$ — non-causal.
- 3-dB bandwidth of $e^{-at}u$: $a$ rad/s. Compress in time ⇔ expand in frequency.

## Key Diagrams to Reproduce




*Fourier transform of the rectangle (Lec 7, sl. 30)*




*$\delta(t) \leftrightarrow 1$ (Lec 7, sl. 43)*